Josephus Queries

Time limit: 1.00 s
Memory limit: 512 MB

CSES 2164 | Mathematics | https://cses.fi/problemset/task/2164

Consider a game where there are n children (numbered 1,2,…,n) in a circle. During the game, every second child is removed from the circle, until there are no children left.

Your task is to process q queries of the form: "when there are n children, who is the kth child that will be removed?"

Input

The first input line has an integer q: the number of queries.

After this, there are q lines that describe the queries. Each line has two integers n and k: the number of children and the position of the child.

Output

Print q integers: the answer for each query.

Constraints

- $1 \le q \le 10^5$
- $1 \le k \le n \le 10^9$

Example

Input

```
4
7 1
7 3
2 2
1337 1313
```

Output

```
2
6
1
1107
```